

Digital Lending Portfolio Optimization

A Data-Driven Framework for Risk-Adjusted Growth





Rapid Digital Lending Growth

- Faster credit approvals
- Technology-driven acquisition
- Expansion into underbanked segments



Rising Portfolio Pressure

- Increasing delinquency risk
- Rising acquisition costs
- Higher borrower volatility



Strategic Challenge

Balancing aggressive growth with sustainable portfolio quality and profitability.

Digital lenders must transition from growth-first scaling to risk-adjusted portfolio optimization.

BUSINESS CHALLENGE



Limited Risk Visibility

Difficulty identifying high-risk borrower segments early in the lending lifecycle.



Delayed Delinquency Detection

Repayment stress indicators are often identified only after portfolio deterioration begins.



Suboptimal Growth Strategies

Aggressive acquisition and scaling strategies may compromise long-term portfolio quality.



Fragmented Leadership Monitoring

Leadership lacks integrated, forward-looking visibility across risk, growth, and profitability metrics.



The institution requires a proactive, data-driven portfolio strategy capable of balancing growth, risk, and profitability simultaneously.



1. Customer Profile

Geography
Employment Type
Bureau Score
Income Proxy



2. Loan & Product Data

Product Type
Loan Amount
Interest Rate
Tenure



3. Repayment Behavior

Payment Timeliness
Delinquency Status
Partial Payments



4. Behavioral Signals

Salary Delays
Spending Volatility
Transaction
Decline Rate



5. Portfolio Outcomes

Default Indicator
Portfolio Quality
Risk-Adjusted
Performance

A synthetic end-to-end lending ecosystem was developed to enable holistic analysis across customer risk, acquisition quality, repayment behavior, and portfolio performance.

PORTFOLIO SNAPSHOT



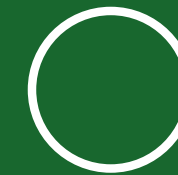
Total Customers
5K

Unique borrowers in the portfolio



Total Loans
9.996K

Loans disbursed across all products



Average Loan Amount
₹287.21K

Average ticket size per loan



Average Interest Rate
19.41%

Weighted average interest rate



Average Days Past Due
8.24

Average delinquency across portfolio



CRO TAKEAWAY

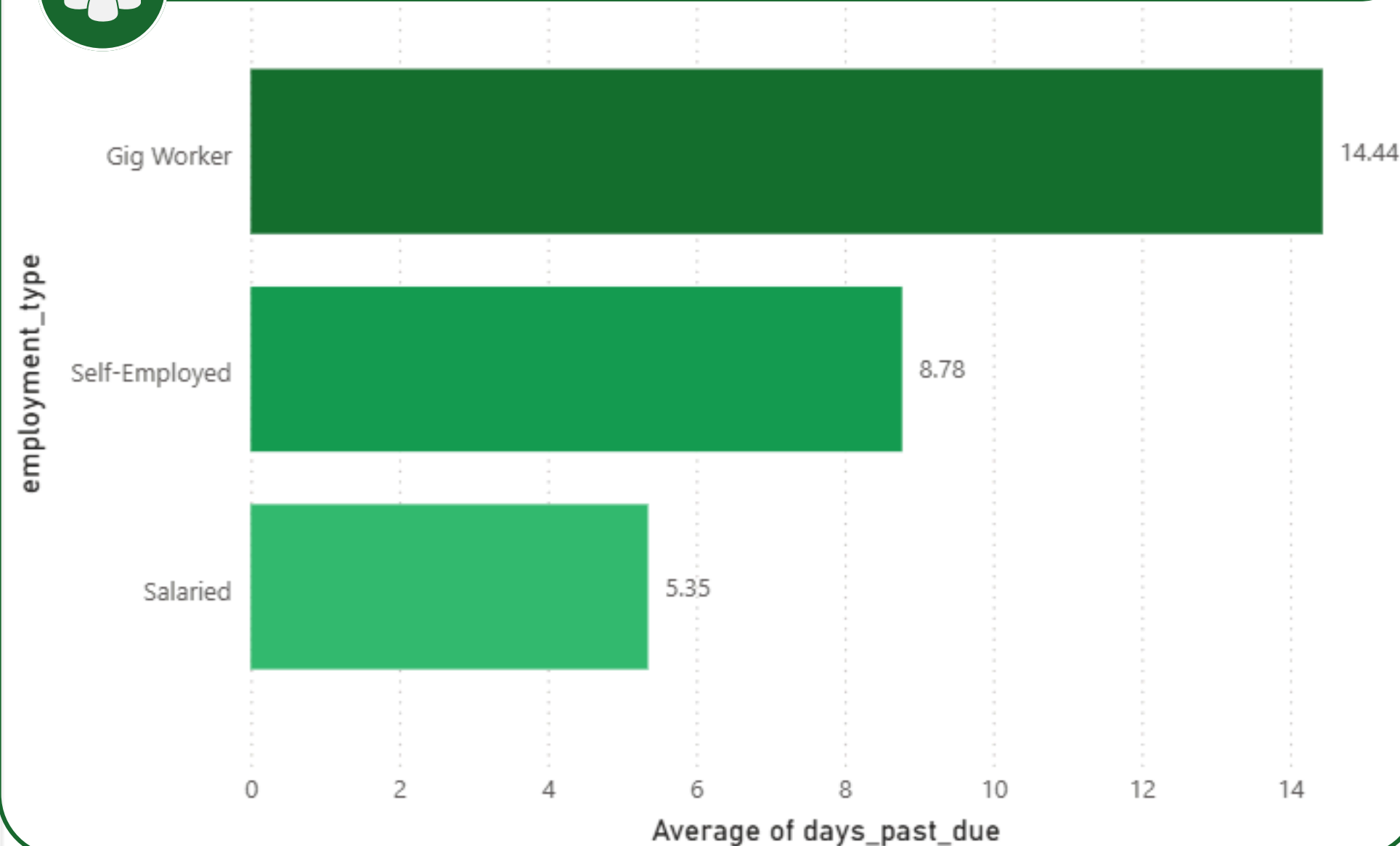
The portfolio demonstrates strong scale; however, delinquency indicators highlight the need for enhanced risk monitoring and proactive portfolio controls.

PORTFOLIO HEALTH STATUS:

MODERATE RISK EXPOSURE

RISK SEGMENTATION ANALYSIS

Delinquency by Employment Type



KEY FINDINGS

Gig-worker and self-employed cohorts demonstrated materially higher delinquency levels relative to salaried borrowers.



- Gig workers exhibited the highest repayment stress indicators.



- Self-employed borrowers showed elevated delinquency volatility.



- Salaried borrowers maintained comparatively stable repayment behavior.



CRO TAKEAWAY

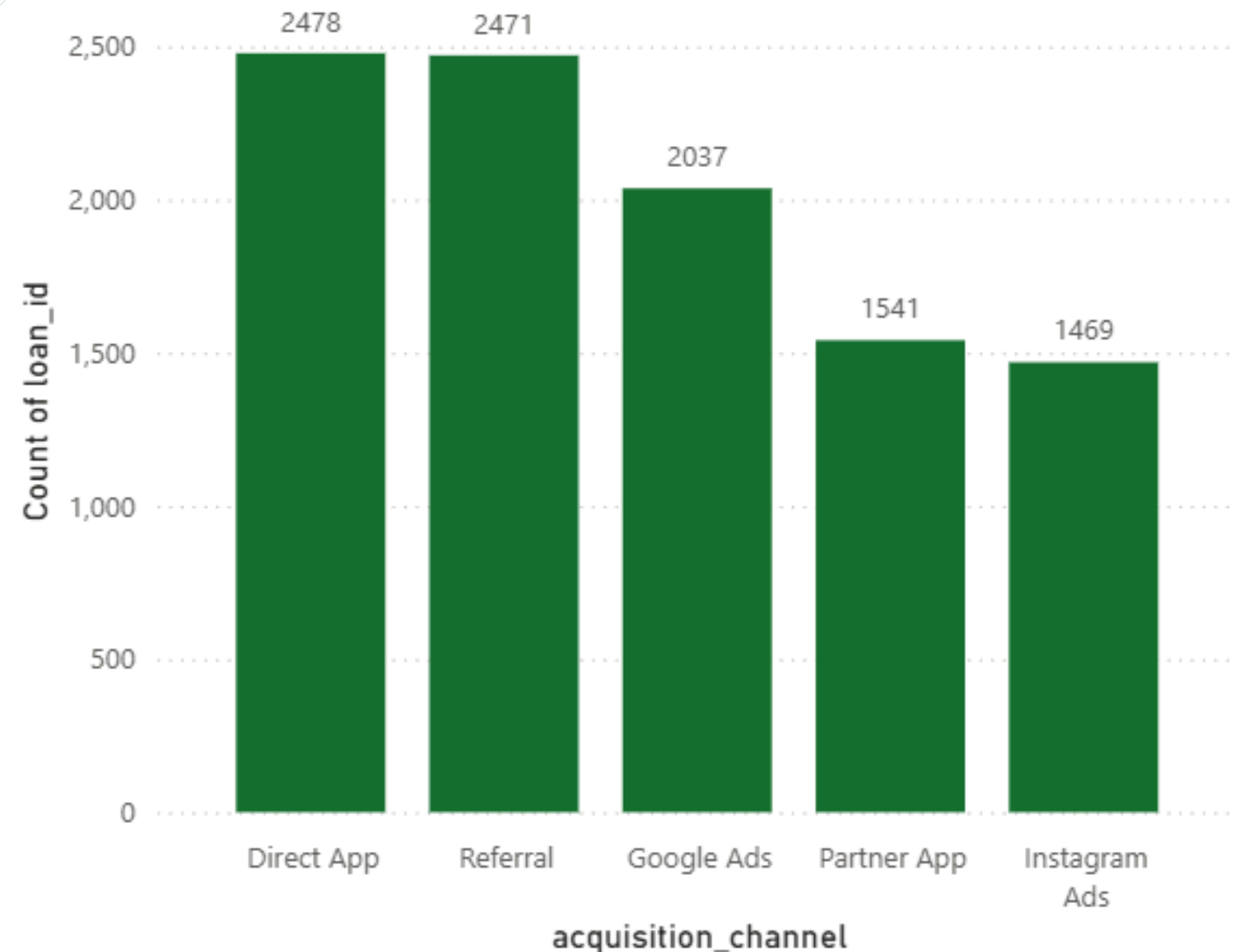
Segment-based underwriting and monitoring frameworks should be strengthened for higher-volatility borrower cohorts.

HIGHER INCOME VOLATILITY → HIGHER DELINQUENCY RISK

ACQUISITION CHANNEL ANALYSIS



Delinquency by Acquisition Channel



KEY FINDINGS

Acquisition-led growth channels demonstrated materially different portfolio quality outcomes across borrower cohorts.



- Direct App and Referral channels generated the highest delinquent loan exposure.



- Paid acquisition channels exhibited elevated portfolio risk relative to controlled sourcing channels.



- Faster customer acquisition may compromise long-term portfolio stability.



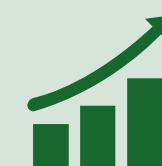
CRO TAKEAWAY

Acquisition performance should be evaluated not only on growth volume, but also on long-term portfolio quality and risk-adjusted returns.



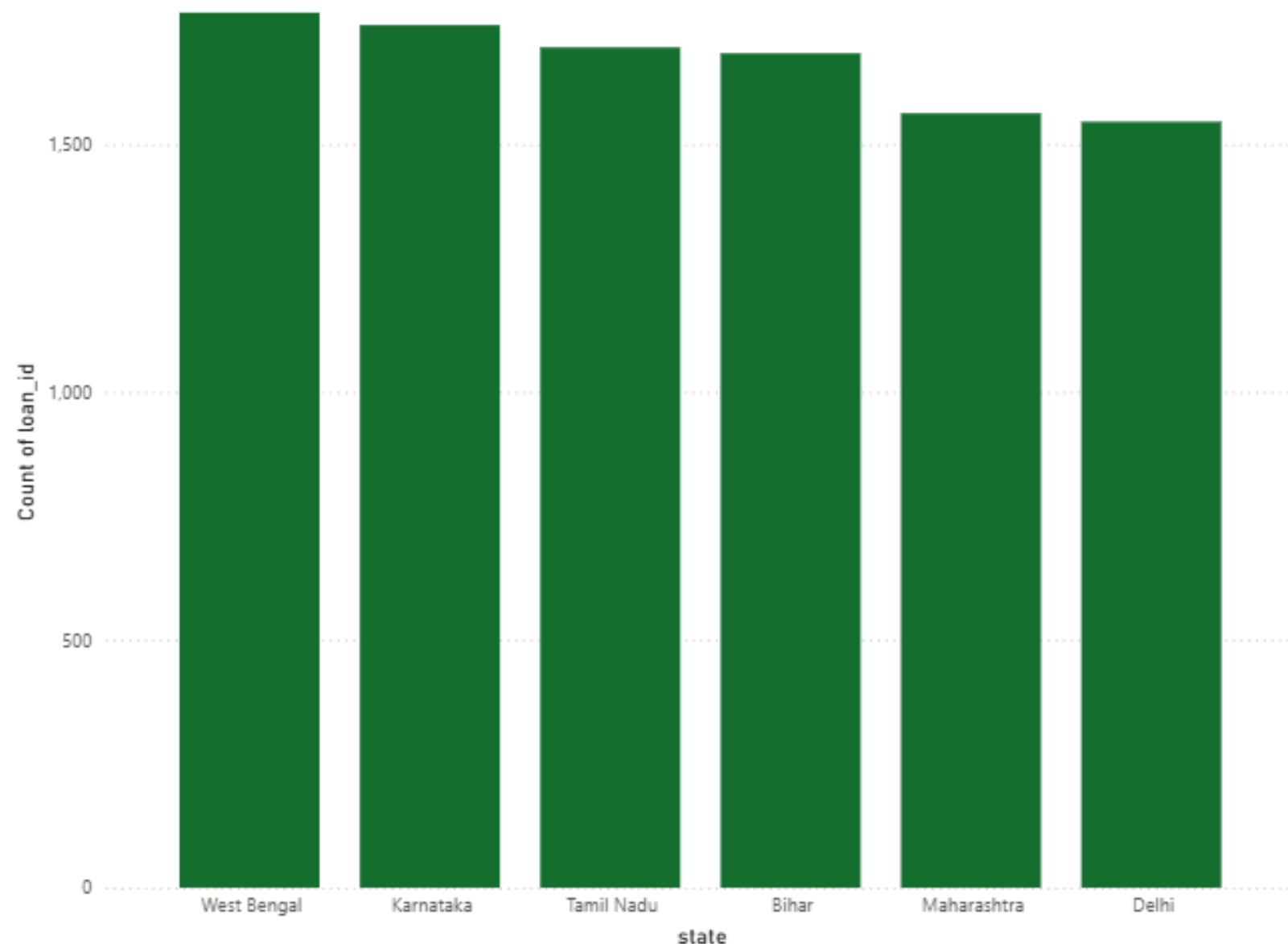
GROWTH EFFICIENCY ≠ PORTFOLIO QUALITY

Faster growth does not always translate to better portfolio outcomes.



GEOGRAPHIC RISK ANALYSIS

Delinquent Loans by State



KEY FINDINGS

Delinquency concentration varies across states, highlighting regional differences in portfolio risk exposure.



West Bengal and Karnataka contributed the highest delinquent loan volumes.



- Larger borrower concentration may increase regional credit stress exposure.



- Geographic diversification is essential for sustainable portfolio expansion.



CRO TAKEAWAY

State-level risk monitoring and regional underwriting controls should be strengthened to manage **concentration risk** effectively.

RISK ISN'T UNIFORM. GROWTH SHOULDN'T BE EITHER.

Data-driven geographic insights enable smarter risk decisions and stronger portfolio resilience.

STRATEGIC RECOMMENDATIONS

Actionable strategies to strengthen portfolio resilience and drive risk-adjusted growth

01



Strengthen Segment-Based Underwriting

Implement differentiated underwriting frameworks for high-volatility borrower cohorts such as gig workers and self-employed applicants.

Outcome: Improve portfolio quality by mitigating higher-risk exposures.

02



Optimize Acquisition Strategy

Prioritize acquisition channels demonstrating stronger long-term portfolio quality and risk-adjusted returns.

Outcome: Enhance ROI by investing in high-quality, low-risk channels.

03



Introduce Geographic Risk Controls

Deploy state-level portfolio monitoring and concentration limits for regions exhibiting elevated delinquency exposure.

Outcome: Reduce regional concentration risk and strengthen geographic diversification.

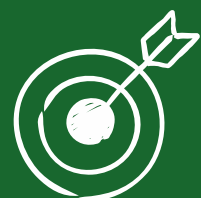
04



Enhance Early Warning Monitoring

Leverage behavioral indicators such as salary delays and repayment stress patterns for proactive delinquency detection.

Outcome: Enable early intervention and minimize credit losses.



Sustainable portfolio growth requires balancing acquisition scale, portfolio quality, and proactive **risk governance** simultaneously.



EXPECTED BUSINESS IMPACT

Strategic outcomes from implementing the recommended risk-driven actions

01



Lower Delinquency Exposure

Enhanced segment-based underwriting can reduce high-risk borrower exposure and improve repayment quality.

Impact: Reduction in delinquency rate and credit losses.

↓ Delinquency Risk

02



Improved Portfolio Quality

Optimized acquisition and geographic diversification strategies strengthen long-term portfolio resilience.

Impact: Stronger asset quality and lower risk concentration.

↑ Portfolio Stability

03



Faster Risk Detection

Behavioral monitoring frameworks enable proactive identification of repayment stress indicators.

Impact: Early intervention reduces delinquencies and improves collections.

↑ Risk Visibility

04

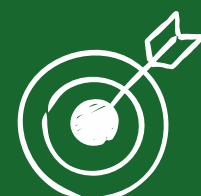


Sustainable Risk-Adjusted Growth

Balanced growth strategies improve profitability while maintaining healthier portfolio performance.

Impact: Higher profitability with sustainable risk-adjusted returns.

↑ Sustainable Growth



A proactive, data-driven lending framework enables **scalable growth** without compromising portfolio quality.



CONCLUSION

Digital lenders must transition from reactive lending models to **proactive, behavior-driven portfolio optimization frameworks.**



DATA-DRIVEN RISK SEGMENTATION

Enables smarter underwriting and improved borrower quality assessment.



BEHAVIORAL MONITORING

Improves early delinquency detection and proactive intervention capability.



SUSTAINABLE GROWTH

Balances acquisition scale with long-term portfolio resilience.



CRO TAKEAWAY

Institutions integrating customer, behavioral, geographic, and acquisition insights into lending strategy will achieve stronger portfolio resilience and sustainable long-term growth.

FROM GROWTH-FIRST LENDING TO RISK-ADJUSTED PORTFOLIO INTELLIGENCE

Thank you very much!

Presented by

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